José Bayoán Santiago Calderón

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Education

2019	PhD Economics	Claremont Graduate University
2014	BA Economics	Southwestern University
	Minors in political science, mathematics, French, and Chinese.	
	Study abroad in Shanghai, China (12FA), and Paris, France (13SP).	

Experience

Sr. Research Economist, U.S. Bureau of Economic Analysis
Suitland, MD · Full-time **2021/05 – Present**

- Published research on national economic accounts and measurement issues, primarily focusing on intellectual property products (IPPs), such as software and data assets.
- Contributed to official statistics through innovative methodologies and operational enhancements. Some examples include the use of big data, natural language processing (NLP), and machine learning, implemented in Python, R, Julia, MATLAB, and Stata.
- Collaborated and engaged with stakeholders, including members of international expert groups and policymakers.

Sr. Scientist II, PumasAI
Remote · Part-time **2020/05 – 2023/11**

Provided strategic and scientific consulting services in model-informed drug development (MIDD), including exposure-response analysis and clinical trial simulations across multiple therapeutic areas, such as rare diseases, metabolic diseases, oncology, and vaccines. Some of the technologies used for the work include Pumas, Julia, R, and SAS.

Postdoctoral Research Associate, University of Virginia
Arlington, VA · Full-time **2019/05 – 2021/05**

- Provided government clients consulting services in the areas of science policy, labor economics, computational models, and program evaluation through problem identification, data discovery (inventory, screening, acquisition), data ingestion, data wrangling (profiling, preparation, linkage, and exploration), fitness-for-use assessment, statistical modeling, and analysis.
- The work employed high-performance computing (HPC) environment and tools like Slurm, Git, containers, and databases (PostgreSQL) to store and access billions of records, some collected through APIs (e.g., REST, GraphQL) that were used in natural language processing (NLP), geographic information systems (GIS), agent-based modeling (ABM), graph theory, and regression analysis using SQL, Julia, and R.

Consultant, JuliaHub
Remote · Part-time **2018/08 – 2019/07**

- Developed components for integrated scientific modeling, simulation, and AI/ML software for quantitative work required by regulatory agencies throughout the drug development cycle. My primary contribution was the development of the bioequivalence module, used in successful FDA submissions.

Data Scientist, Res-Intel
Remote · Full-time **2016/07 – 2018/08**

- Developed the statistical components for the mass-scale residential building energy benchmarking used in demand side management (DSM). The work leveraged various data sources, including real estate information, government administrative records (e.g., building footprint, zoning), weather records, geographic information systems (GIS) layers, and utility meter data. Some of the technologies employed include APIs, databases (SQL), and R.
- Delivered commissioned reports and analyses related to program evaluations of building policies and energy use programs in California.

Research Fellow, Center for Neuroeconomics Studies
Claremont, CA · Part-time **2014/08 – 2016/06**

- Conducted neuroeconomic studies, encompassing aspects of study design, recruitment, experimentation, and data analysis, included in published academic articles.
- Collected and analyzed biometric data, including EDA/GSR, EEG, ECG, and eye tracking, using iMotions.

Selected Peer-Reviewed Publications

J.B. Santiago Calderón and Dylan G. Rassier. 2025. "Valuing the US Data Economy Using Machine Learning and Online Job Postings." Chap. 8 in *Technology, Productivity, and Economic Growth.* University of Chicago Press

G. Korkmaz, **J.B. Santiago Calderón**, B.L. Kramer, L. Guci, and C.A. Robbins. 2024. "From GitHub to GDP: A framework for measuring open source software innovation." *Res. Policy.*

J.B. Santiago Calderón. 2020. "Econometrics.jl." *Proceedings of the JuliaCon Conferences.*

J.B. Santiago Calderón. 2019. "On Cluster Robust Models." PhD diss., Claremont Graduate University

Awards

- U.S. Department of Commerce Silver Medal (2024)
- U.S. Bureau of Economic Analysis Bronze Medal (2023)